CHE 201 HW 11

Due: 11.59pm on 12/07/2019

1. (15 points)

One-tenth kmol of carbon monoxide (CO) in a piston–cylinder assembly undergoes a process from p_1 = 150 kPa, T_1 = 300 Kto p_2 = 500 kPa, T_2 = 370 K. For the process, W = −300 kJ. Employing the ideal gas model, determine

- **a.** the heat transfer, in kJ.
- **b.** the change in entropy, in kJ/K.

Show the process on a sketch of the T–s diagram.

2. (15 points)

Steam enters a turbine operating at steady state at 1 MPa, 200°C and exits at 40°C with a quality of 83%. Stray heat transfer and kinetic and potential energy effects are negligible. Determine (a) the power developed by the turbine, in kJ per kg of steam flowing, (b) the change in specific entropy from inlet to exit, in kJ/K per kg of steam flowing.

3. (15 points)

Water within a piston–cylinder assembly, initially at 10 lbf/in.^{2}, 500°F, undergoes an internally reversible process to 80 lbf/in.^{2}, 800°F, during which the temperature varies linearly with specific entropy. For the water, determine the work and heat transfer, each in Btu/lb. Neglect kinetic and potential energy effects.

4. (15 points)

6.22 Figure P6.22 provides the T–s diagram of a Carnot refrigeration cycle for which the substance is Refrigerant 134a. Determine the coefficient of performance.

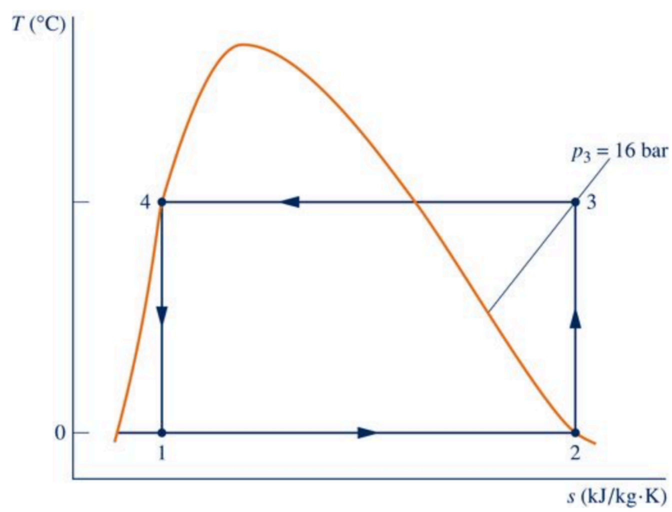

FIGURE P6.22

5. (20 points)

6.40 WP A power plant has a turbogenerator, shown in Fig. P6.40, operating at steady state with an input shaft rotating at 1800 RPM with a torque of 16,700 N · m. The turbogenerator produces current at 230 amp with a voltage of 13,000 V. The rate of heat transfer between the turbogenerator and its surroundings is related to the surface temperature T_b and the lower ambient temperature T_0 and is given by $\dot{Q} = -hA(T_b - T_0)$, where h = 110 W/m² · K, A = 32 m², and T_0 = 298 K.

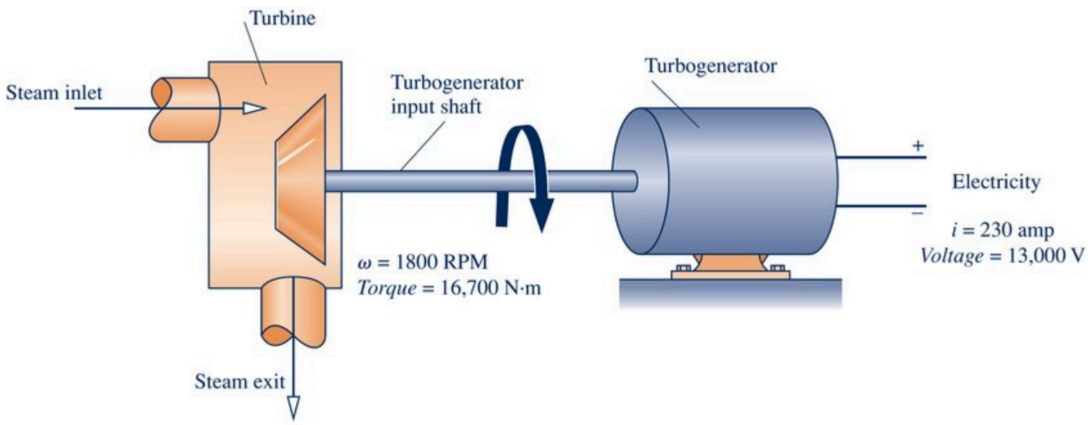

FIGURE P6.40

a. Determine the temperature T_b, in K.

b. For the turbogenerator as the system, determine the rate of entropy production, in kW/K.

c. If the system boundary is located to take in enough of the nearby surroundings for heat transfer to take place at temperature T_0, determine the rate of entropy production, in kW/K, for the enlarged system.

6. (20 points)

6.47 As shown in Fig. P6.47, an insulated box is initially divided into halves by a frictionless, thermally conducting piston. On one side of the piston is 1.0 m³ of air at 400 K, 3 bar. On the other side is 1.0 m³ of air at 400 K, 1.5 bar. The piston is released and equilibrium is attained, with the piston experiencing no change of state. Employing the ideal gas model for the air, determine

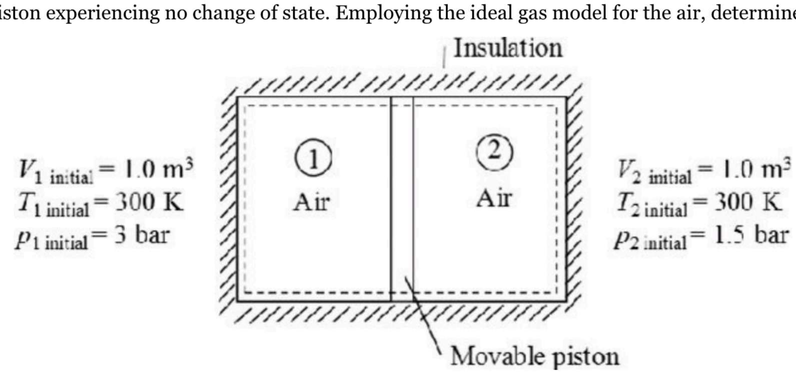

FIGURE P6.47

a. the final temperature of the air, in K.

b. the final pressure of the air, in bar.

c. the amount of entropy produced, in kJ/K.